

Epoch-Dependent Acceleration Scale from Bounded Topology: Predictions for High-Redshift Galactic Dynamics

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Abstract

We propose that the MOND acceleration scale evolves with cosmic epoch as $a_0(z) = a_0(0) E(z)$, where $E(z) = H(z)/H_0$. This single input predicts five correlated high-redshift observables: BTFR normalization (E^{-1}), MOND transition radius ($E^{-1/2}$), asymptotic velocity at fixed baryonic mass ($E^{+1/4}$), Newtonian-inferred lensing mass enhancement ($E^{+1/2}$), and collapse-time shortening ($E^{-1/4}$). The prediction is calibrated to the local SPARC value and compared with the KMOS3D Tully–Fisher offsets, where it faces a 2.9σ trend-shape tension under conservative systematics. Its sharpest near-term test is Euclid DR1 stacked galaxy–galaxy lensing at $z = 0.5$ – 2 : the model predicts a mass- and aperture-independent 74% enhancement of M_{dyn}/M_b at $z = 2$, unlike the mass- and aperture-dependent evolution expected from Λ CDM halo concentration. The topological mechanism fixing $a_0/(cH) = 0.1845$ is presented in Section 2 and Appendix A.

1 Introduction

The MOND acceleration scale $a_0 \approx 1.20 \times 10^{-10} \text{ m/s}^2$ has governed the phenomenology of galactic rotation curves for four decades [1]. Fitted to the local SPARC sample with intrinsic scatter of 0.1 dex [2, 3], it satisfies a numerical coincidence that has never been explained: $a_0/(cH_0) \approx 0.18$, with the Hubble rate setting the dimensional scale of the MOND threshold [4, 5]. Within standard MOND this ratio is an accident; within Λ CDM it has no physical correlate at all. It has remained unexplained since Milgrom first noted it in 1983.

Two current problems motivate examining this scale across cosmic time.

First, the coincidence itself. Using the standard SPARC normalization [2] and Planck 2018 H_0 [6], $a_0/(cH_0) = 0.183$, numerically stable across four decades of improving measurements with no theoretical account of why.

Second, JWST has revealed candidate galaxies at $z \sim 7$ – 10 with stellar masses $M_\star \gtrsim 10^{10} M_\odot$ assembled within 600 Myr of the Big Bang [7, 8]. Under standard halo-growth assumptions, the most extreme candidates imply unusually high star-formation efficiencies and have been discussed as a potential early-structure tension. If a_0 were larger at early times, gravitational collapse would proceed faster and the tension would ease. A related application to the DESI DR2 dark-energy signal [9] is treated in a separate preprint; this paper is limited to the galactic-dynamics consequence of an epoch-dependent a_0 .

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This paper develops one hypothesis and its consequences. Within a bounded-topology measurement framework on $S^1 = \partial(\text{Möbius}) \hookrightarrow S^3$ (Appendix A), the Milgrom ratio resolves to the ratio of two phase-operator values at Fibonacci wells on the 120-domain native to $S^3/2I$:

$$\frac{a_0}{cH} = \frac{C(13/120)}{C(34/120)} = 0.1845, \quad (1.1)$$

agreeing with the observed 0.1833 at the 0.7% level. The ratio itself is not a separate adjustable parameter: once the a_0 and H wells are fixed by separately calibrated well assignments sharing the same edge-mode hierarchy, (1.1) follows algebraically. The non-trivial content is the 0.7% match at a combinatorially sparse position (Section 2). Among the framework’s six Fibonacci-well pairs, (13, 34) is the unique pair that reproduces the observed ratio within 1% (Section 2).

Because both a_0 and H reference the same epoch-dependent hierarchy in the framework’s scaling law, the ratio (1.1) holds at every cosmic epoch. The evolution follows:

$$a_0(z) = a_0(0) E(z), \quad E(z) \equiv H(z)/H_0. \quad (1.2)$$

The bare possibility $a_0 \propto H$ has appeared previously in MOND-motivated discussions of the Milgrom coincidence. The novelty here is not the dimensional guess, but the specific framework mechanism: the ratio is fixed by $C(13/120)/C(34/120)$, the evolution is linear in $E(z)$, and the five observable exponents are locked together rather than tuned independently.

Five testable predictions follow as different powers of $E(z)$, with no additional free parameter beyond the local SPARC calibration. The Euclid DR1 release in October 2026 will deliver the sharpest test through stacked galaxy–galaxy weak lensing at $z = 0.5\text{--}2$. The framework’s monotonic BTFR prediction is currently in tension with the only existing intermediate-redshift measurement [10]; this tension is quantified in Section 4.

Section 2 derives (1.1) and (1.2). Section 3 propagates the evolution into five observable channels. Section 4 treats existing constraints. Section 5 specifies the Euclid test. Section 6 closes.

2 Derivation

The framework’s measurement postulate (Appendix A) maps any Planck-normalized observable to a phase position on the 120-domain native to $S^3/2I$:

$$\frac{A}{A_P} = C(\Theta) \cdot N^n, \quad C(\Theta) = 2 \sin^2(\pi\Theta), \quad (2.1)$$

where $\Theta \in \{k/120\}$ is the phase position, n the manifold-mode index set by the embedding hierarchy $S^1 \subset \text{Möbius} \subset S^3$, and N a dimensionless hierarchy normalization that takes the form $(\sqrt{\Omega})^{-1}$ within each mode class, with its exact value fixed by calibrating one reference observable per sector. The phase operator $C(\Theta)$ is the squared ground-mode intensity on the Möbius surface under anti-periodic boundary conditions (Appendix A.1). A selection rule assigns edge-mode observables ($n = 1$) to the kinematic hierarchy Ω_H and surface-mode observables ($n = 2$) to the eigenvalue hierarchy Ω_Λ (Appendix A.2). The normalization N is associated with the embedding hierarchy: for edge modes ($n = 1$), the relevant kinematic hierarchy is $\Omega_H(z) = (c/(H(z)\ell_P))^2$, giving the leading scale $N_H \sim H(z) t_P$, while the calibrated framework normalization used below is fixed by Eq. (2.3). For surface and space modes, the corresponding normalization is $N_\Lambda = (\sqrt{\Omega_\Lambda})^{-1}$, epoch-independent.

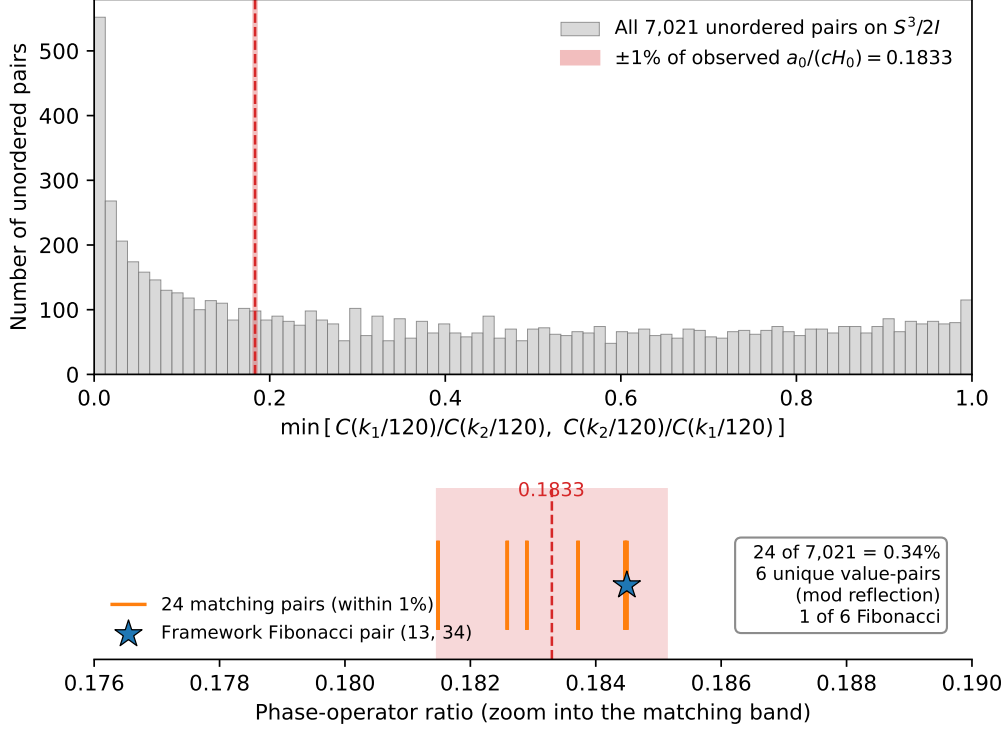


Figure 2.1: Milgrom-ratio sparsity on the 120-domain. Of 7,021 unordered nonzero phase-position pairs, 24 reproduce the observed $a_0/(cH_0) = 0.1833$ within 1%; reflection symmetry $C(k) = C(120 - k)$ collapses these to 6 unique phase-operator value pairs. Within the framework’s Fibonacci-well subset $\{13, 21, 34, 55\}/120$, only the pair $(13, 34)$ matches, marked with a star.

Calibration and prediction. The well assignments calibrated at $z = 0$ (Appendix A.3) place a_0 at $\Theta = 13/120$ and H at $\Theta = 34/120$, both edge modes. The scaling law gives, at any epoch z :

$$\frac{a_0(z)}{a_P} = C(13/120) \cdot N_H(z), \quad (2.2)$$

$$H(z) t_P = C(34/120) \cdot N_H(z). \quad (2.3)$$

Equation (2.3) is the calibration: H is the measured input that fixes $N_H(z) = H(z) t_P / C(34/120)$ at every epoch, with the same role as the QCD coupling measured at a reference process. Equation (2.2) is the prediction: given $N_H(z)$ from (2.3), the well at $\Theta = 13/120$ produces $a_0(z)$. Substituting:

$$\boxed{\frac{a_0(z)}{c H(z)} = \frac{C(13/120)}{C(34/120)} = 0.1845} \quad (2.4)$$

The two well assignments are separately calibrated: H calibrates the edge hierarchy at $\Theta = 34/120$, while a_0 is assigned to $\Theta = 13/120$. Both share the same $N_H(z)$, so (2.4) follows algebraically once the assignments are made. The non-trivial content is not the algebra but the output: the ratio lands at 0.1845 against an observed 0.1833, a 0.7% match at a position where only 0.34% of the domain’s pairs agree within 1%. Numerically, $a_0^{\text{obs}}/(cH_0^{\text{obs}}) = 1.20 \times 10^{-10} / 6.548 \times 10^{-10} = 0.1833$. Of the 7,021 unordered distinct nonzero phase-position pairs on the 120-domain, 24 reproduce this ratio within 1%; the reflection symmetry $C(k) = C(120 - k)$ collapses them to 6 unique phase-operator value pairs. Among the framework’s six Fibonacci-well pairs, $(13, 34)$ is the unique match (Figure 2.1).

Table 3.1: Predicted $a_0(z)$ and deep-MOND enhancement factor across the redshift range relevant to this paper.

z	$E(z)$	$H(z)$ [km/s/Mpc]	$a_0(z)$ [m/s ²]	$\sqrt{E(z)}$	t_{age} [Gyr]
0.0	1.000	67.4	1.20×10^{-10}	1.000	13.79
0.5	1.322	89.1	1.59×10^{-10}	1.150	8.58
1.0	1.791	120.7	2.15×10^{-10}	1.338	5.84
2.0	3.033	204.4	3.64×10^{-10}	1.741	3.27
3.0	4.568	307.9	5.48×10^{-10}	2.137	2.14
5.0	8.297	559.2	9.96×10^{-10}	2.880	1.17
10.0	20.53	1383	2.46×10^{-9}	4.530	0.470
15.0	36.01	2427	4.32×10^{-9}	6.001	0.268

Local-epoch reading. Equation (2.4) holds at every z because $N_H(z)$ is calibrated through the local $H(z)$ at the observer’s epoch, not frozen at $N_H(0)$. The alternative, anchoring N_H to the present, would require a privileged time slice. The bounded domain S^3 with $\partial S^3 = \emptyset$ has no preferred epoch; the standing wave $\Psi(t) = \cos(t/2)$ that governs the cosmic cycle is itself t -dependent throughout. The local-epoch reading is the default; freezing $N_H(0)$ would be the addition of a postulate the topology does not motivate. The two readings are observably distinct at every $z > 0$: at $z = 2$ the BTFR normalization differs by a factor of 3 between them, and the Euclid DR1 test (Section 5) will distinguish them.

Why Λ does not evolve. The same scaling law places Λ at the antinode $\Theta = 60/120$ with $n = 2$ (surface mode), referenced to Ω_Λ : the fixed eigenvalue hierarchy associated with the Λ eigenvalue, not the redshift-dependent fractional density parameter of standard cosmological notation. Under the local-epoch reading, this eigenvalue hierarchy is the same at every z . The phase position $60/120$ has $d \ln C / d\Theta = 0$, giving topological protection against perturbation. The two predictions are structurally inverse: a_0 evolves because it references Ω_H ; Λ does not because it references Ω_Λ . This inversion is forced by the selection rule (Appendix A.2).

Thus the result is a conditional prediction of the framework: given the scaling law, selection rule, calibrated well assignments, and local-epoch reading, the redshift evolution follows by substitution, with no additional evolution parameter.

3 Five observable channels

The evolution $a_0(z) = a_0(0) E(z)$ from Section 2 propagates into galactic observables through different powers of $E(z)$. We adopt the flat Λ CDM Friedmann form $E(z) = \sqrt{\Omega_m(1+z)^3 + \Omega_r(1+z)^4 + \Omega_\Lambda}$ with Planck 2018 parameters [6] ($\Omega_m = 0.315$, $\Omega_r = 9.2 \times 10^{-5}$, $\Omega_\Lambda = 0.685$, $H_0 = 67.4$ km/s/Mpc). The prediction $a_0(z) \propto H(z)$ holds for any expansion history; adopting Λ CDM is a transparency choice that allows direct comparison with published measurements. Table 3.1 provides the reference values consumed by the five channels below.

Table 3.2: BTFR evolution at six reference redshifts.

z	$E(z)$	$A(z)/A(0)$	$v_{\text{flat}}(z)/v_{\text{flat}}(0)$ at fixed M_b
0.0	1.000	1.000	1.000
0.5	1.322	0.756	1.072
1.0	1.791	0.558	1.157
2.0	3.033	0.330	1.320
5.0	8.297	0.121	1.697
10.0	20.53	0.049	2.128

Table 3.3: Predicted MOND radius and asymptotic velocity at $z = 0$ and $z = 2$ for four SPARC archetypes. The fractional shifts $r_M(2)/r_M(0) = 0.574$ and $v_{\text{flat}}(2)/v_{\text{flat}}(0) = 1.320$ apply identically across the table.

Archetype	$M_b [M_\odot]$	$r_M(0)$ [kpc]	$r_M(2)$ [kpc]	$v_{\text{flat}}(0)$ [km/s]	$v_{\text{flat}}(2)$ [km/s]
Dwarf (DDO 154)	3.5×10^8	0.64	0.37	49	64
Sub- L^* (NGC 2403)	1.0×10^{10}	3.41	1.96	112	148
L^* (NGC 6946)	6.0×10^{10}	8.35	4.79	176	232
Giant (UGC 2885)	1.5×10^{11}	13.20	7.58	221	292

3.1 Tully–Fisher normalization: $E(z)^{-1}$

In the deep-MOND limit, the baryonic Tully–Fisher relation (BTFR) is $v_{\text{flat}}^4 = G M_b a_0$ [2]. With evolving a_0 :

$$\boxed{\frac{A_{\text{BTFR}}(z)}{A_{\text{BTFR}}(0)} = \frac{1}{E(z)}}, \quad A_{\text{BTFR}} \equiv \frac{1}{G a_0}. \quad (3.1)$$

The interpolating-function dependence cancels in the ratio: any $\mu(x)$ that fits the local SPARC sample inherits the same fractional evolution. At fixed baryonic mass, $v_{\text{flat}}(z) = v_{\text{flat}}(0) E(z)^{1/4}$; at fixed velocity, $M_b(z) = M_b(0)/E(z)$.

An L^* disk ($M_b = 6 \times 10^{10} M_\odot$, $v_{\text{flat}}(0) = 176$ km/s) is predicted to have $v_{\text{flat}}(z = 2) = 232$ km/s, a 32% increase at fixed baryonic mass. The existing Übler et al. [10] KMOS3D measurement is the only quantitative test of this prediction; the comparison is treated in Section 4.

3.2 MOND transition radius and asymptotic velocity: $E(z)^{-1/2}$, $E(z)^{+1/4}$

The radius where Newtonian gravity equals $a_0(z)$ is $r_M = \sqrt{G M_b / a_0(z)}$, giving

$$\boxed{\frac{r_M(z)}{r_M(0)} = \frac{1}{\sqrt{E(z)}}}. \quad (3.2)$$

At $z = 2$, the transition happens at 57% of the local radius. The asymptotic velocity rises as $E(z)^{1/4}$ (from Section 3.1). Both scalings are mass-independent in the deep-MOND limit.

The comparison isolates the effect of $a_0(z)$ at fixed baryonic distribution. Real galaxies at $z = 2$ differ in structural properties; the observer-facing formulation is: given an observed galaxy at redshift z with measured M_b and $v_{\text{flat}}^{\text{obs}}$, compare $v_{\text{flat}}^{\text{obs}}/E(z)^{1/4}$ to the local SPARC value for galaxies of the same M_b .

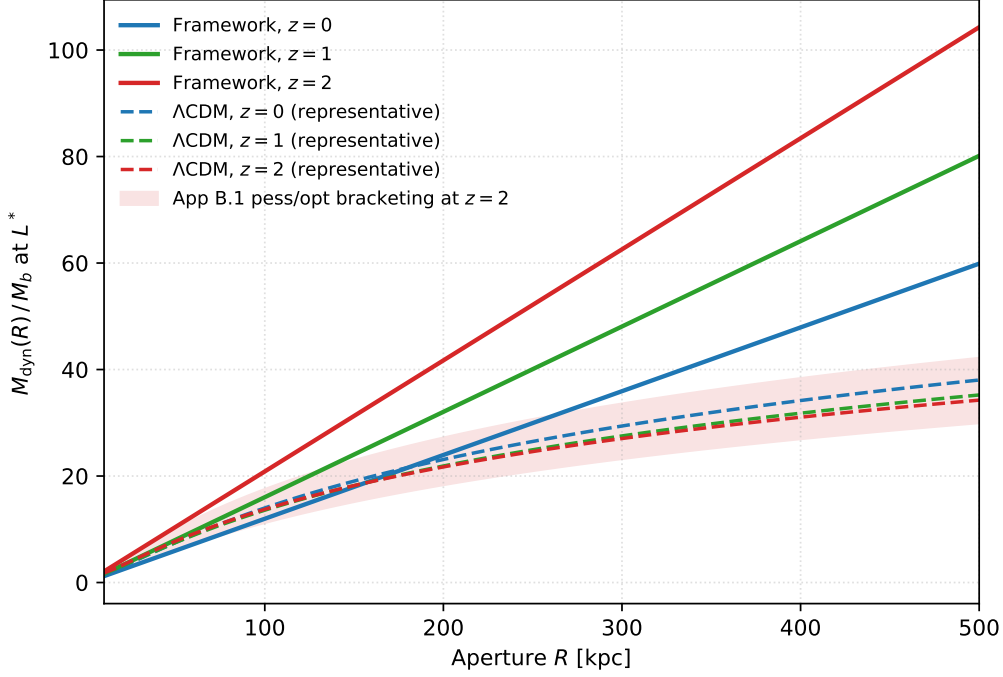


Figure 3.1: Euclid DR1 lensing discriminator. Framework prediction (universal $\sqrt{E(z)}$ enhancement, dashed) versus Λ CDM prediction (mass- and aperture-dependent, solid bands) for M_{dyn}/M_b across galaxy stellar-mass strata at the canonical $R = 100$ kpc aperture. The framework predicts a single 1.74 enhancement factor at $z = 2$ across all masses; Λ CDM spans 37 percentage points across the SPARC archetype range.

Table 3.4: Predicted M_{dyn}/M_b for the L^* archetype at three lensing apertures.

R [kpc]	$z = 0$	$z = 0.5$	$z = 1$	$z = 2$	$z = 5$
30	3.59	4.13	4.81	6.26	10.35
100	11.98	13.77	16.03	20.86	34.50
300	35.93	41.32	48.08	62.57	103.50

3.3 Lensing dynamical mass: $E(z)^{+1/2}$

Galaxy–galaxy weak lensing infers a dynamical mass $M_{\text{dyn}}(R) = v_{\text{flat}}^2 R/G$ within aperture R under the standard Newtonian inversion. The framework’s prediction below applies to this Newtonian-equivalent proxy, not to a relativistic lensing law it does not yet supply: it defines the redshift scaling predicted for the M_{dyn} that lensing analyses report under the standard inversion. In the deep-MOND limit:

$$\frac{M_{\text{dyn}}(R, z)/M_b}{M_{\text{dyn}}(R, 0)/M_b} = \sqrt{E(z)}. \quad (3.3)$$

The enhancement is mass-independent and aperture-independent (for $R \gg r_M$). At $z = 2$, M_{dyn}/M_b is 74% higher than at $z = 0$ at any fixed mass and aperture (Figure 3.1).

Under Λ CDM, halo concentration evolution produces M_{dyn}/M_b shifts that depend on both mass and aperture [11, 12]. The framework predicts a universal $\sqrt{E(z)}$ factor across all masses and apertures. A Euclid DR1 stacked analysis stratified by both galaxy stellar mass and aperture tests the magnitude and the universality of the prediction in a single dataset (Section 5).

Table 3.5: The full predictive content of the evolving acceleration scale, evaluated at $z = 2$.

Observable	Scaling	Value at $z = 2$	Primary test
BTFR normalization	$E(z)^{-1}$	0.330	Euclid DR1, KMOS3D
MOND radius	$E(z)^{-1/2}$	0.574	High- z rotation curves
Asymptotic velocity at fixed M_b	$E(z)^{+1/4}$	1.320	Euclid DR1, KMOS3D
Lensing M_{dyn}/M_b	$E(z)^{+1/2}$	1.741	Euclid DR1 stacked lensing
Free-fall collapse time	$E(z)^{-1/4}$	0.758	JWST spectroscopy

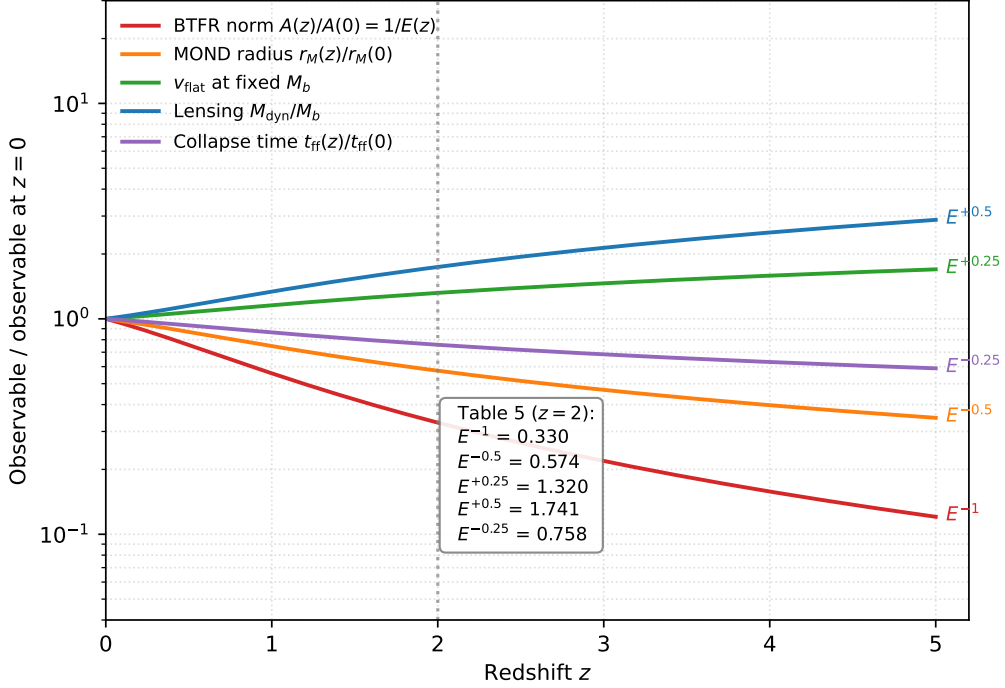


Figure 3.2: Five exponents from one input. Each curve plots the predicted fractional shift in one observable as a function of redshift, derived from the single relation $a_0(z) = a_0(0) E(z)$. Consistency across channels at the same z tests the universality of the deep-MOND scaling.

3.4 Collapse time: $E(z)^{-1/4}$ (supporting evidence)

In the deep-MOND regime, $g_{\text{eff}} = \sqrt{g_N \cdot a_0(z)}$ scales as $\sqrt{E(z)}$ at fixed source, and the free-fall time $t_{\text{ff}} \propto 1/\sqrt{g_{\text{eff}}}$ shortens as $E(z)^{-1/4}$. At $z \sim 8$, the factor $E(z)^{1/4} \approx 2$ shortens the characteristic deep-MOND collapse time by approximately a factor of two. This eases, but does not by itself resolve, the star-formation-efficiency pressure for the Labbé et al. [7] massive candidates. Whether this resolves the impossibility constraint for any specific candidate depends on halo-mass priors the framework does not modify; the per-object analysis is reserved for future work. The fourth-root dependence makes this channel robust: a 10% shift in $E(z)$ changes the correction by less than 3%.

3.5 Summary: five exponents from one input

The five exponents are correlated manifestations of one input (Figure 3.2): any single channel tests $a_0(z) = a_0(0) E(z)$, and agreement among channels at one redshift tests whether the deep-MOND scaling is universal.

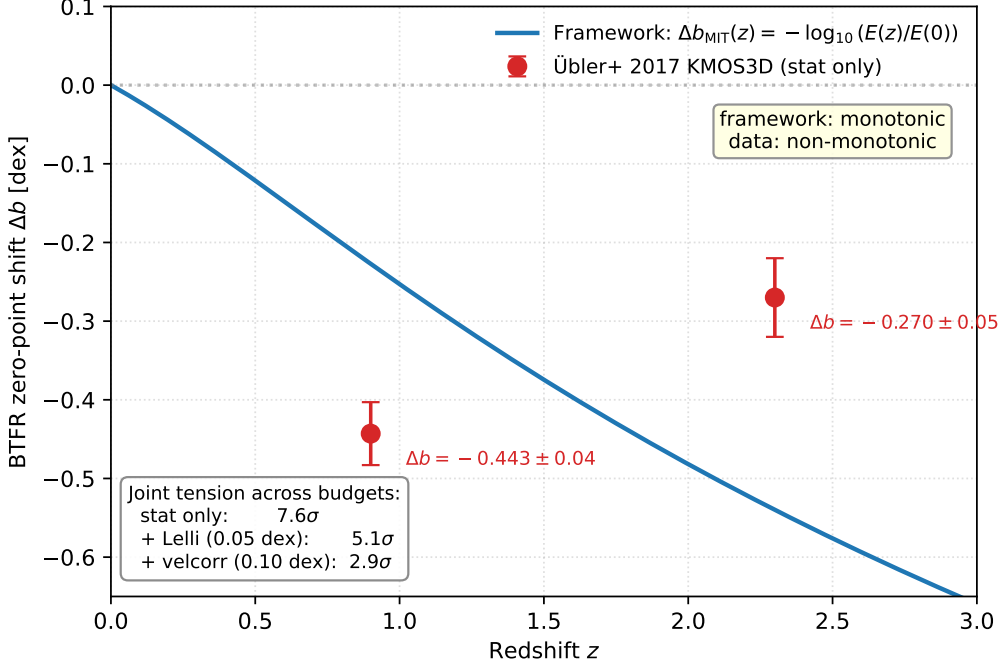


Figure 4.1: KMOS3D BTFR trend-shape tension. The framework predicts a strictly monotonic decrease (dashed); the Übler et al. [10] data are non-monotonic. The joint tension is 2.9σ under the most conservative systematics budget.

Table 4.1: Per-bin and joint σ -tension for the KMOS3D BTFR comparison under three uncertainty budgets.

Budget	$T(z=0.9) [\sigma]$	$T(z=2.3) [\sigma]$	Joint $[\sigma]$
Übler statistical only	-5.4	+5.4	7.6
+ local-baseline (0.05 dex)	-3.4	+3.8	5.1
+ velocity-correction (0.10 dex)	-1.8	+2.2	2.9

4 Existing constraints

4.1 The Übler tension

The KMOS3D analysis of Übler et al. [10] is the only existing quantitative test of the Section 3.1 prediction at intermediate redshift. Their fixed-slope BTFR zero-point offsets relative to the local Lelli et al. [2] baseline are

$$\Delta b^{\text{obs}}(z=0.9) = -0.44 \pm 0.04 \text{ dex}, \quad \Delta b^{\text{obs}}(z=2.3) = -0.27 \pm 0.05 \text{ dex}.$$

The framework predicts

$$\Delta b^{\text{MIT}}(z=0.9) = -0.227 \text{ dex}, \quad \Delta b^{\text{MIT}}(z=2.3) = -0.540 \text{ dex}.$$

The magnitudes overlap: both the prediction and the data place the high- z BTFR substantially below the local relation. The trend shapes do not: the data are non-monotonic ($-0.44 \rightarrow -0.27$), the prediction strictly monotonic ($-0.227 \rightarrow -0.540$) (Figure 4.1).

The tension is quantified across three uncertainty budgets:

Even under the most conservative budget, the joint tension is 2.9σ . We treat this as a provisional tension rather than a formal falsification, because the comparison combines heterogeneous velocity definitions and cross-redshift pressure-support corrections. Formal falsification requires a matched-systematics measurement: single instrument, uniform tracer, identical correction prescription across redshift bins (Section 5).

We tested whether the Übler radial pressure-support correction can produce the observed non-monotonic pattern from a monotonic underlying BTFR. Mock galaxies at $z = 0.9$ and $z = 2.3$ were generated under the framework’s prediction with literature-realistic distributions of baryonic mass, scale length, and velocity dispersion [13]. The Übler thick-disk correction $v_{\text{circ}}^2(r) = v_{\text{rot}}^2(r) + 2\sigma_0^2 r/R_d$ was applied with the published selection cut $v_{\text{rot,max}}/\sigma_0 > 4.4$. Four bias models (radial-position uncertainty, beam-smearing of σ_0 , non-asymptotic rotation curves, selection-induced mass shifts) were swept over literature-plausible ranges. No combination reproduces the observed $(-0.44, -0.27)$ pattern. Pulling biases to optimize the $z = 0.9$ fit gives $\Delta b(0.9) = -0.29$ dex, leaving the $z = 0.9$ point 0.15 dex above observed; at the same parameters $\Delta b(2.3) = -0.65$ dex, 0.38 dex below observed. The joint-L2 best fit gives $(\Delta b(0.9), \Delta b(2.3)) = (-0.17, -0.43)$ at residuals $(+0.28, -0.16)$ dex. The mock-generation script, bias parameterization, and random seeds are archived with the rest of the analysis pipeline at the Zenodo deposit cited in the Data availability section.

The pattern is not a velocity-correction artifact. The prediction stands. Resolution will come from Euclid DR1 stacked lensing and JWST/ground-IFU matched-tracer follow-up at the original Übler redshifts (Section 5).

4.2 Galaxy clusters

MOND analyses of local galaxy clusters [14] report a residual mass discrepancy of roughly a factor of 5 at $r \approx 0.5 R_{\text{vir}}$. The framework’s evolving a_0 does not address this: the relevant clusters sit at $z \lesssim 0.15$, where $E(z) \leq 1.08$ and the framework correction is less than 8%. A factor-of-5 problem is not touched by an 8% correction. This discrepancy is an open problem shared by all MOND-class theories [3, 15, 16] and predates the present framework, which neither introduces nor resolves it.

4.3 The CMB at recombination

A naive application of $a_0(z)$ to cosmological perturbations at $z = 1090$ gives $a_0 \approx 23,000 \times a_0(0)$, placing recombination-scale perturbations deep in the MOND regime and disrupting the acoustic peak structure that Planck fits at sub-percent precision.

The framework’s selection rule (Section 2, Appendix A.2) separates this channel structurally. The rule assigns a_0 to the edge sector ($n = 1, \Omega_H$) and cosmological perturbations to the space sector ($n = 3, \Omega_\Lambda$). Under this assignment, the $a_0(z)$ scaling governs galactic dynamics and does not propagate to perturbation evolution. The same rule independently produces the Section 2 Milgrom-ratio match at 0.7% and the Λ -constant prediction; it was not introduced to address the CMB.

The consistency condition is quantitative. Writing a minimal leakage coupling $g_{\text{eff}} = g_N + \varepsilon \sqrt{g_N a_0(z)}$, the Planck first-peak amplitude, measured to 0.5% precision, constrains $\varepsilon \leq 1.2 \times 10^{-5}$. The framework predicts $\varepsilon = 0$ under exact edge/space decoupling, satisfying the bound by five orders of magnitude. A first-principles perturbation-theory derivation of the decoupling is the principal open task for the framework’s CMB-scale predictions.

Table 4.2: Constraint status across five regimes.

Regime	Status
Local ($z = 0$)	Consistent by construction
Intermediate- z BTFR (Übler [10])	2.9σ tension on trend shape
Galaxy clusters [14]	Inherited, unaddressed ($\sim 8\%$ vs factor-5)
CMB ($z = 1090$)	Structurally decoupled; $\varepsilon \leq 1.2 \times 10^{-5}$ from Planck
Strong-lens time delays	Below current sensitivity

4.4 Other regimes

Local SPARC and THINGS measurements are consistent by construction: the well assignments were calibrated against $a_0(0)$ and H_0 at $z = 0$ (Section 2). The Genzel et al. [22] SINS/zC-SINF sample of six massive star-forming disks at $z = 0.9$ – 2.4 reports declining outer rotation curves with low inferred dark-matter fractions; because no fixed-slope BTFR zero-point is published, the data are qualitatively consistent with Section 3.1 but do not provide a quantitative test of the normalization prediction. Strong-lensing time-delay cosmography (TDCOSMO [17], H0LiCOW [18]) is below current sensitivity to the framework’s predictions. None of these regimes is sharp enough to test the framework’s normalization prediction independently.

4.5 Summary

One live tension, one inherited problem, one structural decoupling, two consistencies. The Übler tension is held against the prediction without retreat.

5 The Euclid DR1 test

Euclid’s stated objective is to characterize the dark Universe and its evolution across $z = 0$ – 2 [19]. The present paper focuses on the galactic-dynamics prediction $a_0(z) \propto H(z)$. The Euclid DR1 release in October 2026 is expected to deliver the sharpest test of this prediction through stacked galaxy–galaxy weak lensing at $z = 0.5$ – 2 .

What Euclid measures. The wide survey is expected to provide the photometric redshifts, stellar-mass estimates, galaxy shapes, and sample sizes needed for a stacked galaxy–galaxy lensing test. For the framework, the relevant observable is the Newtonian-equivalent M_{dyn}/M_b inferred from stacked tangential shear profiles after mass and redshift stratification. Equation (3.3) is a prediction for this quoted proxy, not a standalone relativistic lensing law.

What the framework predicts. At fixed baryonic mass and aperture, M_{dyn}/M_b is enhanced by $\sqrt{E(z)}$ relative to $z = 0$: a 15% excess at $z = 0.5$, 34% at $z = 1$, and 74% at $z = 2$ (Table 3.4). The enhancement is universal across galaxy mass and lensing aperture in the deep-MOND regime.

Why this discriminates from Λ CDM. Under Λ CDM, M_{dyn}/M_b also evolves with redshift through halo concentration evolution and stellar-to-halo mass relation shifts [11, 12]. The Λ CDM prediction is mass-dependent and aperture-dependent: at $z = 2$ relative to $z = 0$, the Λ CDM ratio shifts by a factor of 0.76 for a Sub- L^* galaxy, 0.98 for L^* , and 1.13 for a Giant at the canonical $R = 100$ kpc aperture, spanning 37 percentage points across the mass range

Table 5.1: Falsification criteria for the five Section 3 predictions. A matched-systematics discrepancy at $\geq 2\sigma$ is treated as a reportable tension; formal falsification requires either $\geq 3\sigma$ in one channel or mutually consistent $\geq 2\sigma$ failures across independent redshift bins or observables.

Prediction	Signature	Reportable tension if	Key systematic
BTFR normalization	$A(z)/A(0) = 1/E(z)$	Single- z inconsistency at $\geq 2\sigma$	Pressure-support correction
BTFR trend shape	Monotonic decrease	Non-monotonic trend confirmed under matched systematics	Cross-instrument tracer mismatch
MOND radius	$r_M(z)/r_M(0) = E(z)^{-1/2}$	Single- z inconsistency at $\geq 2\sigma$	Baryonic mass model
Lensing enhancement	$\sqrt{E(z)}$, mass/aperture-independent	Inconsistency at $\geq 2\sigma$, or mass/aperture-dependent shift	Halo concentration evolution
Collapse-time correction	$t_{\text{ff}}(z)/t_{\text{ff}}(0) = E(z)^{-1/4}$	Corrected formation timescales incompatible with spectroscopic ages at $\geq 2\sigma$	Halo-mass priors

Table 5.2: Near-term test schedule.

Window	Instrument	Tests	Delivery
Stacked lensing, $z = 0.5\text{--}2$	Euclid DR1	Lensing enhancement, universality	October 2026
Emission-line sample selection	Euclid NISP grism	Lens sample definition, redshift binning	October 2026
Matched-tracer BTFR	JWST NIRSpec IFU / ground IFU	BTFR normalization, trend shape	In progress
Resolved kinematics at $\ddot{\text{U}}$ bler redshifts	JWST NIRSpec IFU	BTFR at $z = 0.9$, $z = 2.3$	In progress
Spectroscopic confirmation, $z = 7\text{--}9$	JWST NIRSpec	JWST efficiency	2025–2026

(Appendix B). The framework predicts the same factor 1.74 for all three. A Euclid DR1 analysis stratified by both stellar mass and aperture tests the magnitude and the universality of the prediction in a single dataset.

Falsification criteria. The lensing row is the cleanest because $\sqrt{E(z)}$ is universal while the Λ CDM alternative is not. A mass- and aperture-stratified Euclid analysis tests both the magnitude of the prediction and its universality simultaneously.

All programs listed are existing or imminent. No new instrument or observational capability is required. This paper and its predictions are deposited on Zenodo (concept DOI: [10.5281/zenodo.19980665](https://doi.org/10.5281/zenodo.19980665)) prior to the Euclid DR1 release, fixing the numbers before the data arrive.

6 Conclusions

The Milgrom coincidence $a_0 \approx cH_0$ is accounted for, within the bounded-topology framework of Appendix A, by a fixed ratio of two phase-operator values at Fibonacci wells: $a_0/(cH) = C(13/120)/C(34/120) = 0.1845$, agreeing with the observed 0.1833 at 0.7%. The a_0 and H wells are fixed by separately calibrated assignments at $z = 0$ sharing the same edge-mode hierarchy; the ratio follows algebraically and is unique among the framework’s Fibonacci-well pairs (Section 2).

Because both observables reference the same epoch-dependent hierarchy, the ratio holds at every cosmic epoch: $a_0(z) = a_0(0) E(z)$. Five observable channels follow as different powers of $E(z)$, with no additional free parameter. The BTFR normalization shifts as $E(z)^{-1}$, the MOND transition radius contracts as $E(z)^{-1/2}$, the asymptotic velocity at fixed baryonic mass rises as $E(z)^{+1/4}$, the lensing-inferred dynamical mass enhancement scales as $E(z)^{+1/2}$, and the gravitational collapse time shortens as $E(z)^{-1/4}$. The five exponents move together, set by one input; cross-channel agreement at a fixed redshift is the test of the deep-MOND scaling.

The prediction lives in a constraint landscape with one tension (Übler et al. KMOS3D BTFR trend shape at 2.9σ , tested by forward modeling), one inherited problem (cluster-scale MOND discrepancy, untouched by the framework’s $\lesssim 8\%$ low-redshift correction), and one structural decoupling (CMB via the selection rule, with leakage bounded to $\varepsilon \leq 1.2 \times 10^{-5}$ by Planck). The Übler tension is acknowledged, not explained away.

A separate preprint applies the same selection rule to the dark-energy sector (DOI: [10.5281/zenodo.19798852](https://doi.org/10.5281/zenodo.19798852)); the present paper is limited to the galactic-dynamics consequence $a_0(z) = a_0(0) E(z)$.

The sharpest scheduled test is Euclid DR1 in October 2026: stacked galaxy–galaxy lensing at $z = 0.5\text{--}2$ stratified by mass and aperture, probing both the predicted 74% enhancement and its universality in a regime where the framework and Λ CDM produce quantitatively distinct shifts. This paper and its predictions are deposited on Zenodo (concept DOI: [10.5281/zenodo.19980665](https://doi.org/10.5281/zenodo.19980665)) prior to that release.

A Framework foundations

This appendix supplies the structural detail behind the Section 2 derivation. The bounded topology, phase operator, scaling law, and selection rule are stated in Section 2. This appendix develops the 120-domain (A.1), the selection rule’s physical basis (A.2), and the well-assignment eligibility conditions (A.3).

A.1 The 120-domain

The physical observable space is the quotient $S^3/2I$, where $2I$ is the binary icosahedral group with $|2I| = 120$. The discrete subgroups of $SU(2) \cong S^3$ are classified: cyclic groups $\mathbb{Z}_n$, binary dihedral groups $2D_n$, and three exceptional groups (binary tetrahedral $|2T| = 24$, binary octahedral $|2O| = 48$, binary icosahedral $|2I| = 120$). Two constraints select $2I$.

The quotient $S^3/2I$ is the Poincaré homology sphere: the classical example of a closed 3-manifold with the integral homology of S^3 but nontrivial fundamental group, obtained as the quotient of S^3 by the binary icosahedral group $2I$ [23]. The same group occupies the E_8 position in the ADE classification of finite subgroups of $SU(2)$, and its representation theory maps to the E_8 Dynkin diagram through the McKay correspondence [24]. More generally, quotients S^3/Γ by finite freely acting subgroups are spherical space forms, classified in the standard space-form literature [25]. The framework’s use of $2I$ is therefore not ad hoc: within the finite exceptional binary polyhedral subgroups of $SU(2)$, it is the largest case and the E_8 endpoint of the ADE/McKay classification.

First, $2I$ is the largest exceptional discrete subgroup of $SU(2)$, giving the maximum spectral resolution compatible with S^3 . In the framework, the quotient order $|2I| = 120$ defines the discrete phase domain used to label observable positions, with spacing $\Delta\Theta = 1/120$.

Second, the icosahedron is the unique Platonic solid whose branch orders $(2, 3, 5)$ are consecutive Fibonacci numbers satisfying $2 + 3 = 5$. This Fibonacci-recurrence structure makes

the Fibonacci positions on the 120-domain the natural stable lattice (A.2 below and the Hurwitz stability argument in A.3).

The phase position is quantized: $\Theta \in \{k/120 : k = 0, 1, \dots, 119\}$. For photon-mediated observables, the bosonic projection $|\psi|^2$ erases the anti-periodic sign flip, halving the resolution to a 60-position grid (even numerators only). The framework implements this as a bosonic projection: photon-mediated observables sample the 60-position quotient, while dynamical observables may access the full 120-domain.

The chronon, the smallest phase advance the domain can register, is $\Delta t_{\min} = 4\pi/120 = \pi/30$, giving a minimum action $\Delta S_{\min} = \hbar\pi/30$ that is frame-independent by construction.

A.2 The selection rule

The scaling law (Section 2, Eq. 2.1) assigns each observable a manifold-mode index n from the embedding hierarchy $S^1 \subset \text{Möbius} \subset S^3$ and a corresponding hierarchy normalization N . The physical basis for the (n, Ω) assignment is:

Edge modes ($n = 1, \Omega_H$). The boundary S^1 is a kinematic locus: it carries no intrinsic eigenvalue (the surface eigenvalue lives one dimension up, on the Möbius strip, where Λ is placed). What an edge-mode observable references is the embedding scale of S^1 in the ambient cosmological geometry. The natural such scale is the Hubble horizon $R_H = c/H$, and its dimensionless Planck ratio is $\Omega_H = (c/(H\ell_P))^2$. Because H evolves with cosmic epoch, edge-mode observables inherit that evolution through the calibrated normalization $N_H(z) = H(z)t_P/C(34/120)$.

Surface modes ($n = 2, \Omega_\Lambda$). The Möbius strip carries an intrinsic eigenvalue: the ground mode of the Laplace–Beltrami operator on the totally geodesic surface in S^3 gives $\lambda_0 = 2/R^2$, where R is the curvature radius of S^3 . The Gauss–Codazzi embedding converts this to $\Lambda_{\text{obs}} = (3/2)\lambda_0 = 3/R^2$ under three conditions: totally geodesic embedding ($K_{ij} = 0$), isotropy (CMB-verified to 10^{-5}), and de Sitter vacuum. The hierarchy ratio $\Omega_\Lambda = (\Lambda\ell_P^2)^{-1}$ is set by this eigenvalue and does not evolve.

Space modes ($n = 3, \Omega_\Lambda$). Cosmological perturbations propagate on the spatial slice S^3 and reference the same fixed eigenvalue hierarchy as the surface. Under this assignment, perturbation evolution is governed by Ω_Λ , not Ω_H , and inherits no $a_0(z)$ modification. This is the structural basis for the CMB consistency argument in Section 4.3.

A.3 Well assignments

Three eligibility conditions determine which Fibonacci wells are consistent with each observable:

1. **Manifold index** separates edge modes ($n = 1: H_0, a_0$) from surface modes ($n = 2: \Lambda$).
2. **Bosonic projection.** Photon-mediated observables access only the 60-grid (even numerators); dynamical observables access the full 120-grid.
3. **Antinode requirement.** Λ , as a topologically protected eigenvalue, sits at $\Theta = 60/120$ where $d \ln C/d\Theta = 0$.

The Fibonacci wells on the 120-domain are $\{13, 21, 34, 55\}/120$, corresponding to F_7 through F_{10} . The lower bound $F_7 = 13$ is the noise floor: below this, the amplitude $C(k/120)$ is

indistinguishable from neighboring positions. The upper bound $F_{10} = 55$ is set by the reflection symmetry $C(k) = C(120 - k)$, which makes $F_{11} = 89$ equivalent to $F_8 = 21$. These bounds are motivated by the Hurwitz stability criterion: φ is the irrational hardest to approximate by rationals of bounded denominator, so Fibonacci positions on the 120-domain are taken to minimize destructive interference between modes.

Under the eligibility conditions:

- (a) a_0 is assigned $\Theta = 13/120$: the unique coprime position ($\gcd(13, 120) = 1$), accessible only on the full 120-grid as required for a dynamical (non-photon-mediated) observable.
- (b) H_0 is assigned $\Theta = 34/120$: the unique even-numerator Fibonacci well among $13/120$, $21/120$, $34/120$, and $55/120$, consistent with the bosonic-projection condition for a photon-mediated observation.
- (c) Λ is assigned $\Theta = 60/120$: the antinode, where the phase operator takes its maximum and the log-slope vanishes.

The wells $21/120$ and $55/120$ remain unassigned in this paper. The assignments have the status of empirical calibrations at $z = 0$ that land on positions compatible with the eligibility conditions. The manifold-mode classification, bosonic-projection filter, selection rule, and the prediction $a_0(z) \propto H(z)$ were established in the framework’s foundational deposit [20] prior to the present paper’s calibration, foreclosing post-hoc adjustment of the eligibility conditions to fit the well assignments. A first-principles derivation from the Hurwitz/Fibonacci structure of the 120-domain is future work.

B Λ CDM lensing comparison methodology

Section 3.3 and Section 5 compare the framework’s universal $\sqrt{E(z)}$ lensing enhancement against the Λ CDM prediction under halo concentration evolution. This appendix specifies the parameterization.

B.1 Galaxy archetypes and halo parameters

Four baryonic-mass archetypes span the SPARC range:

Table B.1: Baryonic-mass archetypes spanning the SPARC range.

Archetype	$M_b [M_\odot]$
Dwarf (DDO 154)	3.5×10^8
Sub- L^* (NGC 2403)	1.0×10^{10}
L^* (NGC 6946)	6.0×10^{10}
Giant (UGC 2885)	1.5×10^{11}

Representative halo masses and concentrations at the L^* archetype, consistent with the Moster et al. [11] stellar-to-halo mass relation and the Duffy et al. [12] concentration evolution:

Sub- L^* and Giant archetypes use mass-appropriate (M_{halo}, c) values from the same parameterization. The SHMR mapping is used only to set representative halo masses; the discriminator depends on the relative mass- and aperture-dependence of the Λ CDM prediction, not the absolute baryonic-to-halo normalization. These are representative defaults; alternative SHMR inversions

Table B.2: Representative halo masses and concentrations at the L^* archetype.

z	$M_{\text{halo}} [M_{\odot}]$	c
0	1.5×10^{12}	7.5
1	1.0×10^{12}	5.0
2	7.0×10^{11}	3.5

shift the absolute Λ CDM values but preserve the mass- and aperture-dependence that constitutes the Section 5 discriminator.

B.2 NFW enclosed mass

The dynamical mass at fixed physical aperture R is

$$M_{\text{dyn}}(R, z) = M_{\text{NFW}}(R; M_{\text{halo}}, c, z) + M_b,$$

with NFW enclosed mass

$$M_{\text{NFW}}(R) = M_{\text{halo}} \cdot \frac{\ln(1 + R/r_s) - (R/r_s)/(1 + R/r_s)}{\ln(1 + c) - c/(1 + c)},$$

scale radius $r_s = R_{\text{vir}}/c$, and virial radius under the R_{200} convention:

$$R_{\text{vir}}^3 = \frac{3 M_{\text{halo}}}{4\pi \cdot 200 \cdot \rho_{\text{crit}}(0) E(z)^2}.$$

The ratio M_{dyn}/M_b is computed under the same Newtonian inversion used by the framework prediction (Section 3.3, Eq. 3.3).

B.3 The discriminator

At the canonical $R = 100$ kpc aperture, the Λ CDM M_{dyn}/M_b ratio at $z = 2$ relative to $z = 0$ is:

Table B.3: Λ CDM M_{dyn}/M_b ratio at $z = 2$ relative to $z = 0$, versus the framework’s universal prediction.

Archetype	Λ CDM ratio ($z=2/z=0$)	Framework ratio
Sub- L^*	0.76	1.74
L^*	0.98	1.74
Giant	1.13	1.74

The Λ CDM prediction spans 37 percentage points across the mass range (0.76 to 1.13); the framework predicts a universal 1.74. At fixed L^* mass, the Λ CDM ratio also varies with aperture: 1.07 at $R = 30$ kpc, 0.98 at $R = 100$ kpc, 0.92 at $R = 300$ kpc; the framework predicts 1.74 at all three.

Sensitivity bracketing. Table B.4 evaluates the same NFW pipeline under pessimistic and optimistic SHMR + concentration parameterizations at the L^* archetype, $R = 100$ kpc.

Even under the pessimistic parameterization, the framework’s $z = 2$ prediction sits 19% above the Λ CDM upper bracket. The normalized evolution-ratio framing ($1.74/0.98 \simeq 1.78$)

Table B.4: Λ CDM $L^* M_{\text{dyn}}/M_b$ at $R = 100$ kpc under three parameterizations, against the framework’s universal prediction.

Parameterization	(M_{halo}, c) at $z = 0$	(M_{halo}, c) at $z = 2$	M_{dyn}/M_b $z = 0 / z = 2$	Framework / Λ CDM at $z = 2$
Pessimistic	$(2.0 \times 10^{12}, 9.0)$	$(1.0 \times 10^{12}, 4.5)$	17.7 / 17.5	1.19
Representative	$(1.5 \times 10^{12}, 7.5)$	$(7.0 \times 10^{11}, 3.5)$	14.0 / 13.8	1.52
Optimistic	$(1.0 \times 10^{12}, 6.0)$	$(5.0 \times 10^{11}, 2.5)$	10.3 / 11.2	1.86
Framework (universal)	—	—	11.98 / 20.86	—

inherits this robustness with additional cancellation, since the Λ CDM $z = 2$ and $z = 0$ endpoints shift correlatedly under SHMR and concentration variations and the ratio is more stable than either absolute value.

A Euclid DR1 stacked analysis stratified by both stellar mass and aperture tests the framework on two axes simultaneously: the magnitude of the redshift shift and its universality across mass and aperture. In normalized redshift evolution at L^* , the discriminator is $1.74/0.98 \simeq 1.78$, well above the statistical scale expected for large stacked samples, though the limiting uncertainty will be systematic calibration rather than raw shape noise.

Cosmological parameters: Planck 2018 [6], $\rho_{\text{crit}}(0) = 8.53 \times 10^{-27}$ kg/m³, $\Omega_m = 0.315$, $\Omega_\Lambda = 0.685$, $H_0 = 67.4$ km/s/Mpc.

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Data availability

All numerical predictions, tabulated framework outputs, and source code for the analysis pipelines and figure-generation scripts are archived at Zenodo (concept DOI: [10.5281/zenodo.19980665](https://doi.org/10.5281/zenodo.19980665)), mirrored from <https://github.com/dmobius3/a0z>. The archive includes the combinatorial baseline table (supporting Section 2); the Sections 3–5 prediction tables (Tables 3.1–3.5) in CSV format; the Λ CDM lensing discriminator overlay (NFW enclosed-mass curves at the canonical archetypes, supporting Section 3.3 and Appendix B); the Übler σ -tension table and the four-bias forward-model script (mock-generation, bias parameterization, and random seeds; supporting Section 4.1); the CMB leakage-bound table (supporting Section 4.3); and the Labbé candidate speedup factors (supporting Section 3.4).

Conflict of interest

The author declares no conflict of interest.

References

- [1] M. Milgrom. A modification of the Newtonian dynamics as a possible alternative to the hidden mass hypothesis. *Astrophys. J.*, 270:365, 1983. doi:[10.1086/161130](https://doi.org/10.1086/161130).
- [2] F. Lelli, S. S. McGaugh, and J. M. Schombert. SPARC: Mass models for 175 disk galaxies with Spitzer photometry and accurate rotation curves. *Astron. J.*, 152:157, 2016. doi:[10.3847/0004-6256/152/6/157](https://doi.org/10.3847/0004-6256/152/6/157).
- [3] B. Famaey and S. S. McGaugh. Modified Newtonian Dynamics (MOND): Observational phenomenology and relativistic extensions. *Living Rev. Relativ.*, 15:10, 2012. doi:[10.12942/lrr-2012-10](https://doi.org/10.12942/lrr-2012-10).
- [4] M. Milgrom. The modified dynamics as a vacuum effect. *Phys. Lett. A*, 253:273, 1999. doi:[10.1016/S0375-9601\(99\)00077-8](https://doi.org/10.1016/S0375-9601(99)00077-8).
- [5] S. S. McGaugh. The mass discrepancy–acceleration relation: Disk mass and the dark matter distribution. *Astrophys. J.*, 611:26, 2004. doi:[10.1086/421901](https://doi.org/10.1086/421901).
- [6] Planck Collaboration. Planck 2018 results. VI. Cosmological parameters. *Astron. Astrophys.*, 641:A6, 2020. doi:[10.1051/0004-6361/201833910](https://doi.org/10.1051/0004-6361/201833910).
- [7] I. Labbé et al. A population of red candidate massive galaxies ~ 600 Myr after the Big Bang. *Nature*, 616:266, 2023. doi:[10.1038/s41586-023-05786-2](https://doi.org/10.1038/s41586-023-05786-2).
- [8] M. Boylan-Kolchin. Stress testing Λ CDM with high-redshift galaxy candidates. *Nat. Astron.*, 7:731, 2023. doi:[10.1038/s41550-023-01937-7](https://doi.org/10.1038/s41550-023-01937-7).
- [9] DESI Collaboration. DESI DR2 results. II. Measurements of baryon acoustic oscillations and cosmological constraints. arXiv:2503.14738, 2025.
- [10] H. Übler et al. The evolution of the Tully–Fisher relation between $z \sim 2.3$ and $z \sim 0.9$ with KMOS^{3D}. *Astrophys. J.*, 842:121, 2017. doi:[10.3847/1538-4357/aa733b](https://doi.org/10.3847/1538-4357/aa733b).
- [11] B. P. Moster, T. Naab, and S. D. M. White. Galactic star formation and accretion histories from matching galaxies to dark matter haloes. *Mon. Not. R. Astron. Soc.*, 428:3121, 2013. doi:[10.1093/mnras/sts261](https://doi.org/10.1093/mnras/sts261).
- [12] A. R. Duffy, J. Schaye, S. T. Kay, and C. Dalla Vecchia. Dark matter halo concentrations in the Wilkinson Microwave Anisotropy Probe year 5 cosmology. *Mon. Not. R. Astron. Soc.*, 390:L64, 2008. doi:[10.1111/j.1745-3933.2008.00537.x](https://doi.org/10.1111/j.1745-3933.2008.00537.x).
- [13] E. Wisnioski et al. The KMOS^{3D} survey: Design, first results, and the evolution of galaxy kinematics from $0.7 < z < 2.7$. *Astrophys. J.*, 799:209, 2015. doi:[10.1088/0004-637X/799/2/209](https://doi.org/10.1088/0004-637X/799/2/209).
- [14] E. Pointecouteau and J. Silk. New constraints on modified Newtonian dynamics from galaxy clusters. *Mon. Not. R. Astron. Soc.*, 364:654, 2005. doi:[10.1111/j.1365-2966.2005.09590.x](https://doi.org/10.1111/j.1365-2966.2005.09590.x).
- [15] R. H. Sanders. Clusters of galaxies with modified Newtonian dynamics. *Mon. Not. R. Astron. Soc.*, 342:901, 2003. doi:[10.1046/j.1365-8711.2003.06596.x](https://doi.org/10.1046/j.1365-8711.2003.06596.x).

- [16] G. W. Angus, B. Famaey, and D. A. Buote. X-ray group and cluster mass profiles in MOND: Unexplained mass on the group scale. *Mon. Not. R. Astron. Soc.*, 387:1470, 2008. doi:[10.1111/j.1365-2966.2008.13353.x](https://doi.org/10.1111/j.1365-2966.2008.13353.x).
- [17] M. Millon et al. TDCOSMO: I. An exploration of systematic uncertainties in the inference of H_0 from time-delay cosmography. *Astron. Astrophys.*, 639:A101, 2020. doi:[10.1051/0004-6361/201937351](https://doi.org/10.1051/0004-6361/201937351).
- [18] K. C. Wong et al. H0LiCOW XIII: A 2.4 per cent measurement of H_0 from lensed quasars: 5.3 σ tension between early- and late-Universe probes. *Mon. Not. R. Astron. Soc.*, 498:1420, 2020. doi:[10.1093/mnras/stz3094](https://doi.org/10.1093/mnras/stz3094).
- [19] Euclid Collaboration: Y. Mellier et al. Euclid: I. Overview of the Euclid mission. arXiv:2405.13491, 2024.
- [20] B. Shatto. Mode Identity Theory: Modal realization from nested topology. Zenodo, 2025. doi:[10.5281/zenodo.18064856](https://doi.org/10.5281/zenodo.18064856).
- [21] S. S. McGaugh, F. Lelli, and J. M. Schombert. Radial acceleration relation in rotationally supported galaxies. *Phys. Rev. Lett.*, 117:201101, 2016. doi:[10.1103/PhysRevLett.117.201101](https://doi.org/10.1103/PhysRevLett.117.201101).
- [22] R. Genzel et al. Strongly baryon-dominated disk galaxies at the peak of galaxy formation ten billion years ago. *Nature*, 543:397, 2017. doi:[10.1038/nature21685](https://doi.org/10.1038/nature21685).
- [23] N. Saveliev. *Lectures on the Topology of 3-Manifolds: An Introduction to the Casson Invariant*. de Gruyter, Berlin, 2nd edition, 2012.
- [24] J. McKay. Graphs, singularities, and finite groups. In *Proceedings of Symposia in Pure Mathematics*, volume 37, page 183. American Mathematical Society, 1980.
- [25] J. A. Wolf. *Spaces of Constant Curvature*. AMS Chelsea Publishing, Providence, RI, 6th edition, 2011.